Ultimate Nutrition

Creapure Capsules

To perform optimally, muscles need energy. The body provides the energy needed in situations requiring immediate, high-intensity actions as in exercise, in the form of ATP, or adenosine triphosphate. Since the body has only a limited supply of ATP, usually to last only a few seconds of intense exercise, ATP is continuously produced to supply energy in order for the muscles to function. The burst of energy is produced by the breakdown of ATP when one phosphate group is released, which packs considerable metabolic energy. The body uses creatine phosphate to quickly replenish ATP.

The more energy the muscles store, the better they can perform in events, which require intense, immediate energy, such as weightlifting, sprinting, jumping, football, hockey and soccer. Since creatine is stored in the muscle as creatine phosphate, intake of supplemental creatine can increase the production of energy enabling muscles to perform at higher intensity. While the body produces its own supply of creatine, it is not sufficient to supply the muscle with the added energy necessary for intense performance.

The benefits of creatine supplementation for endurance athletes have been actively researched. This research has established that creatine can, in fact, extend endurance at a relatively high dose of 20 grams per day. Creatine increases the muscle mass and muscle girth if taken along with a sustained exercise regimen. Initially, it may also increase weight due to gain in the muscle mass, which may slow down some people, especially swimmers. The "slowing down" may be due to the highly aerobic nature of this exercise, and should be reversible after sustained exercise.

The ergogenic, or performance-enhancing, effect of creatine is best achieved by creatine monohydrate. Ideally, the increase of creatine in the body is achieved by a five- to seven-day "loading" period followed by a maintenance period. Since more stored creatine will produce more energy, it is best to optimize its uptake into the muscle during the maintenance phase by supplementing it with carbohydrates. Carbohydrates increase creatine uptake into muscle and reduce its excretion in the urine.

Creatine is a power- and performance-enhancing nutrient, and should be ingested with an exercise regimen in place. By helping the muscle release energy during exercise, it increases the muscle mass and strength.

SELECTED REFERENCES

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FREQUENTLY ASKED QUESTIONS

1 What is the Creatine Dosage During the Loading Phase?
A total of 20 to 25 grams per day of creatine are usually recommended during the loading phase. Since this is a large amount to be taken in one sitting, it should be broken down into four or five servings of around 5 grams each, which is equivalent to 5-6 capsules.

2 How Should Creatine Be Taken?
Creatine is best ingested with a nutritious fluid in generous amounts.

3 What Are the Side Effects of Creatine?
Creatine does not have any adverse side effects per se if taken in optimal doses tailored to a reasonable exercise regimen. In large amounts, it can, however, cause gastrointestinal distress. If this situation presents itself, the intake of creatin should be reduced to individual comfort levels.

4 Doesn't Creatine Initially Decrease Performance?
Not necessarily! Creatine intake, however, does increase weight because the muscle gains mass. Therefore, in intensely aerobic exercises, such as swimming and trail running, an athlete's performance may not be up to the par. This can be reversed, however, with increasing the endurance training, which is also stimulated by creatine over time.

5 How Does Creatine Compare with Androstenedione?
Both creatine and androstenedione are taken by athletes to help build bigger, stronger muscles. Androstenedione and creatine have some similarities but they also differ from each other considerably.

Androstenedione and creatine are both found in the body. Whereas androstenedione is a prohormone, creatine is an amino acid that the body uses to generate energy for muscle function.

Androstenedione is a steroid hormone used to raise testosterone levels which, in turn, stimulates the production of proteins that are the building blocks of muscle, bone and other lean body tissues. Creatine, on the other hand, is not a steroid, but its anabolic properties are being increasingly understood. Creatine helps make more "fuel" available to the muscle in the form of ATP. This extra fuel ensures that the muscle works faster and longer and recovers faster after intense exercise or sustained physical activity. This also means that there is less muscle soreness and enhanced endurance. Numerous studies support the use of creatine for increasing endurance, building muscle mass and improving athletic performance.